ALGONQUIN COLLEGE | Centre for Continuing & Online Learning

Assignment - Data Transform Activity (20%)

Due: Module 4

Purpose:

The purpose of this assignment is to transform data into an appropriate format for analysis and then extract that data for further problem solving.

Assignment Objective:

For the business problem you selected, you will extract data for further analysis using a series of transformations, including: select, mutate, filter, and arrange.

Instructions:

This activity relates to the Video Games Sales Data:

Click on the link below to download your data:

- **Video Games Sales - Data**

In this assignment, you will perform simple transformations on your data (you might need to reload the data in HDInsight and Hive). Do the following steps:

1) a) Create a new table for your analysis call “sales_genre”.
   b) Load the table “sales” into this table.
   c) Select these columns: Genre, Global_Sales, Critic_Score.

2) Round the data found in the “Global_Sales” column. (HINT: the SQL function to round a number is ROUND(obs)

3) Filter the data to only look at those items in the “Critic_Score” column that are greater than 0.

4) Order the data in the “Critic_Score” column from highest to lowest. (HINT: Use the DESC query)

This assignment relates to the following Course Learning Requirements:

CLR 3: Recommend which tools and the best use of these tools, to support data analysis, data quality, problem solving, analytics and business decision machining.

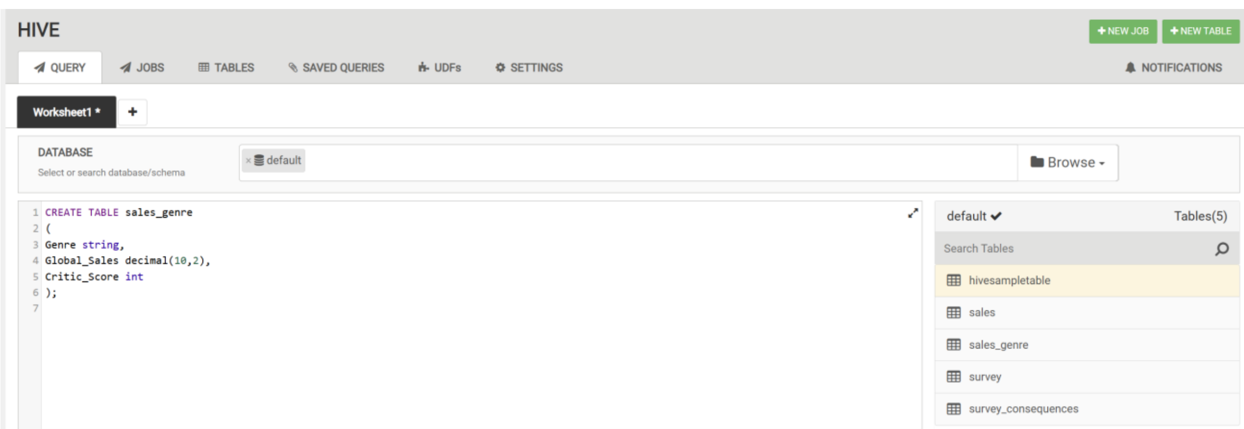
CLR 4: Extract and load data to support problem solving and decision-making.

Grading for This Assignment

Steps required	Grade Earned
1. a) Create a new table for your analysis call “sales_genre”. b) Load the table “sales” into this table. c) Select these columns: Genre, Global_Sales, Critic_Score. 2. Round the data found in the “Global_Sales” column. (HINT: the SQL function to round a number is ROUND(obs)	50%
3. Filter the data to only look at those items in the “Critic_Score” column that are greater than 0.	80%
4. Order the data in the “Critic_Score” column from highest to lowest. (HINT: Use the DESC query)	100%

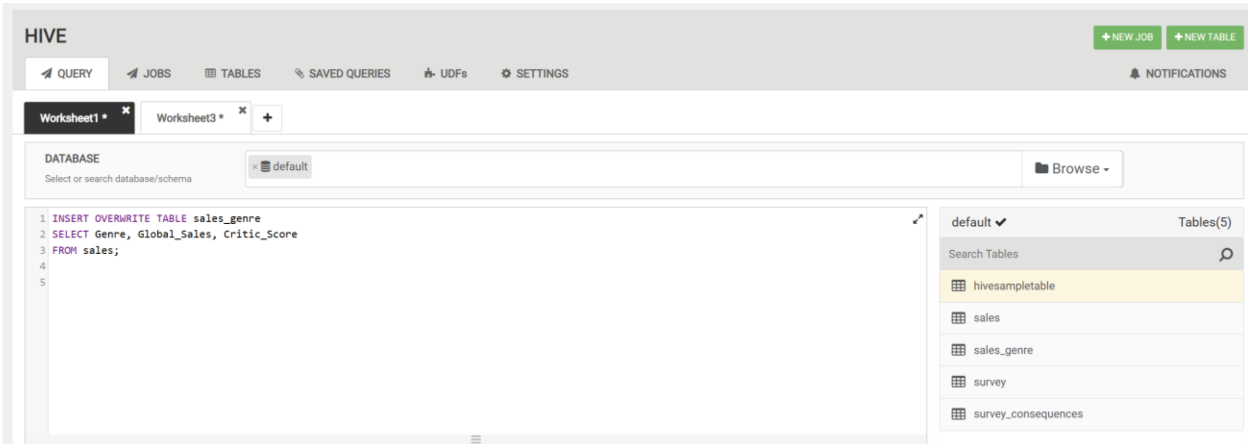
1A)

```
CREATE TABLE sales_genre
(
Genre string,
Global_Sales decimal(10,2),
Critic_Score int
);
```

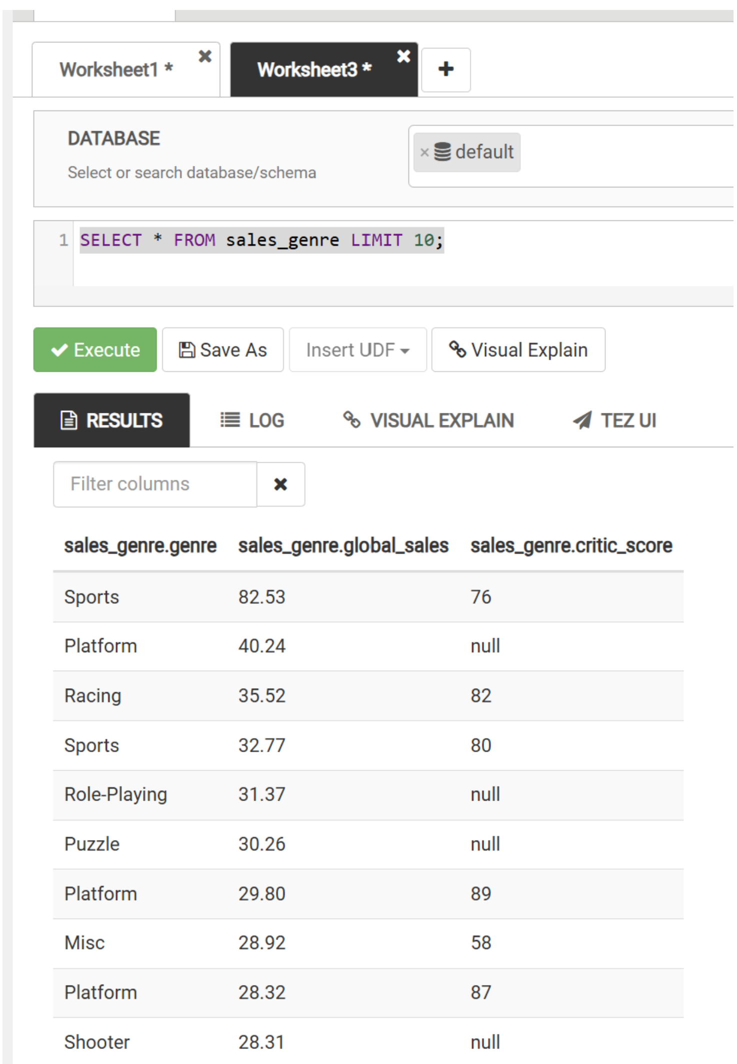


1B) & 1C)

```
INSERT OVERWRITE TABLE sales_genre
SELECT Genre, Global_Sales, Critic_Score
FROM sales;
```



RESULTS FROM:
SELECT * FROM sales_genre LIMIT 10;



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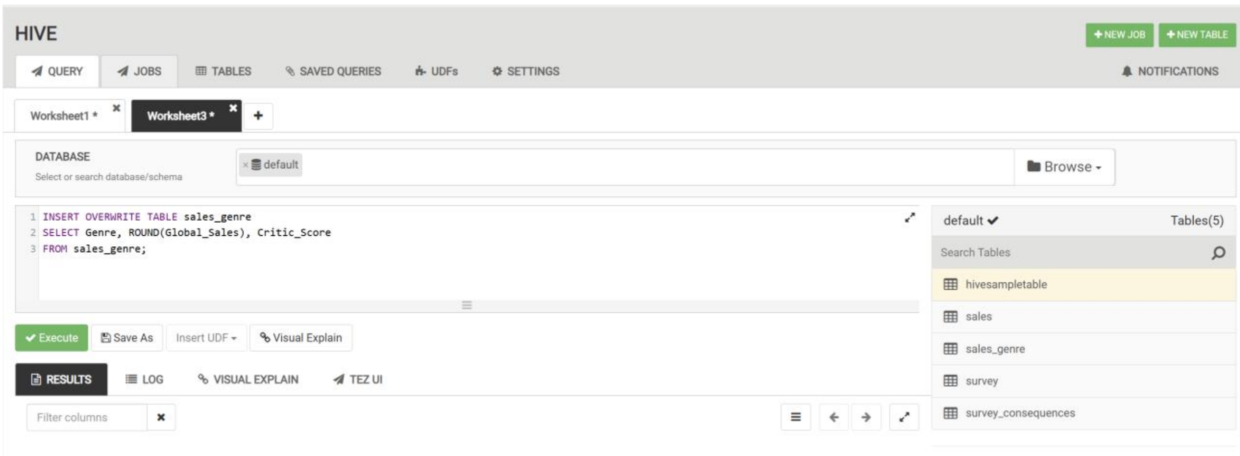
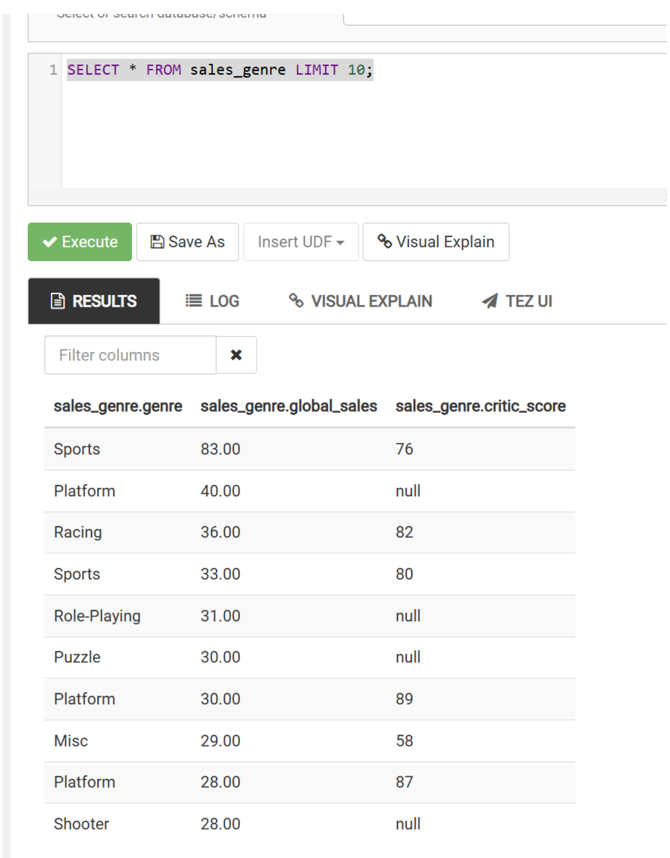
2)

INSERT OVERWRITE TABLE sales_genre
SELECT Genre, ROUND(Global_Sales), Critic_Score
FROM sales_genre;

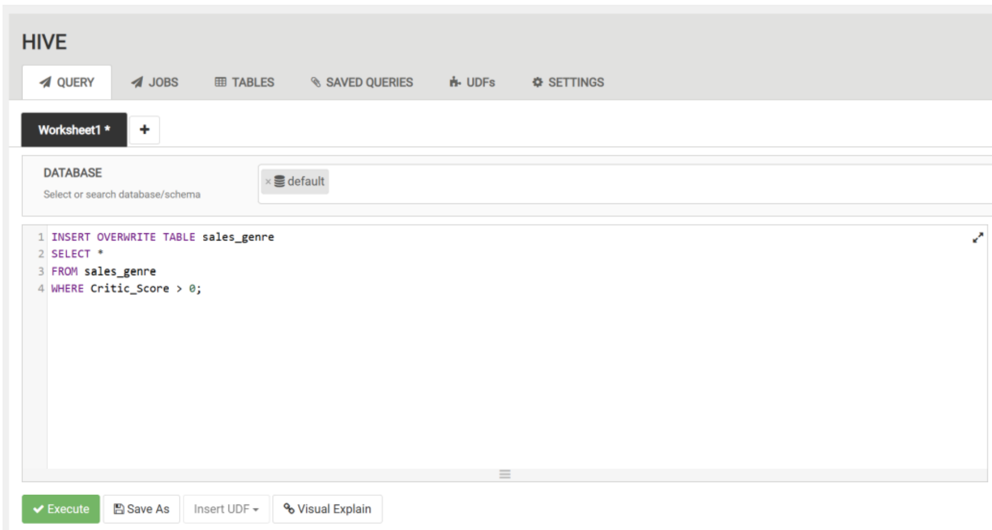


RESULTS FROM:
SELECT * FROM sales_genre LIMIT 10;

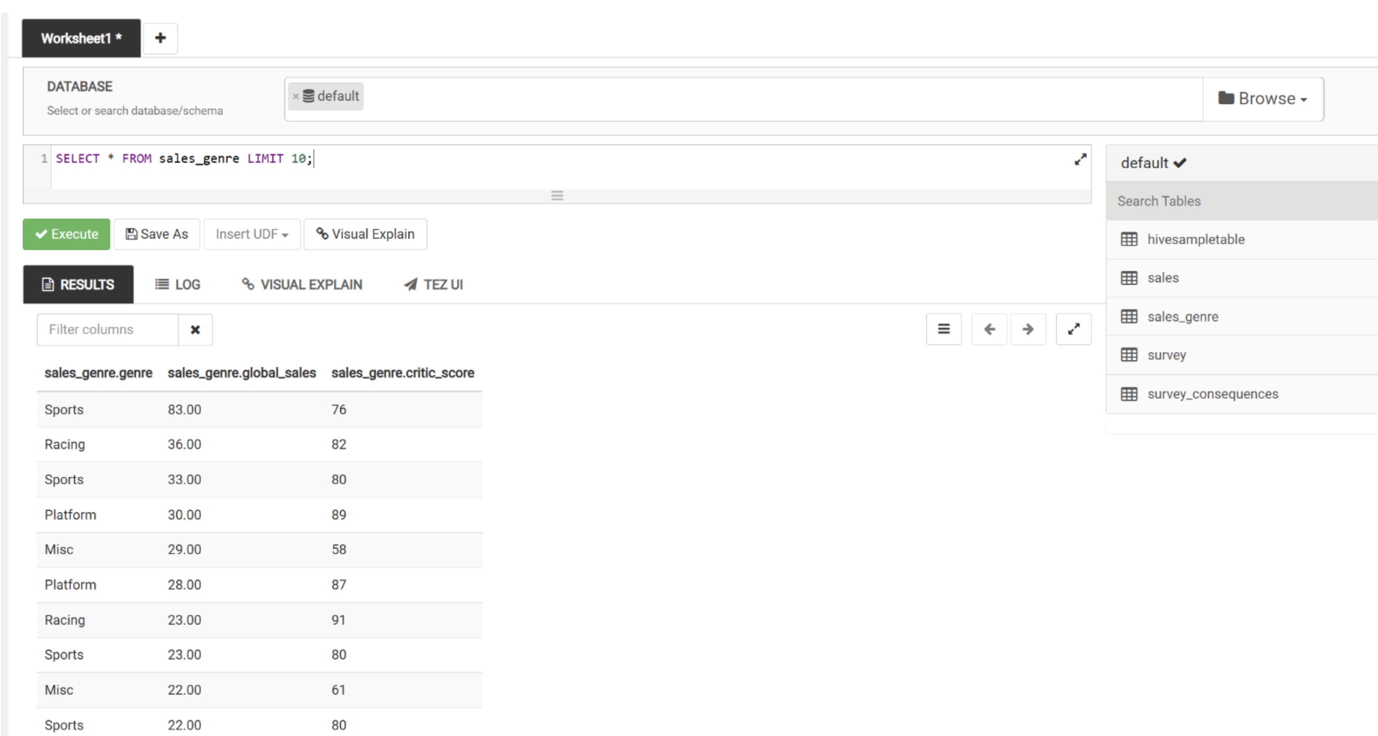


3)

INSERT OVERWRITE TABLE sales_genre
SELECT *
FROM sales_genre
WHERE Critic_Score > 0;



RESULTS FROM:
SELECT * FROM sales_genre LIMIT 10;



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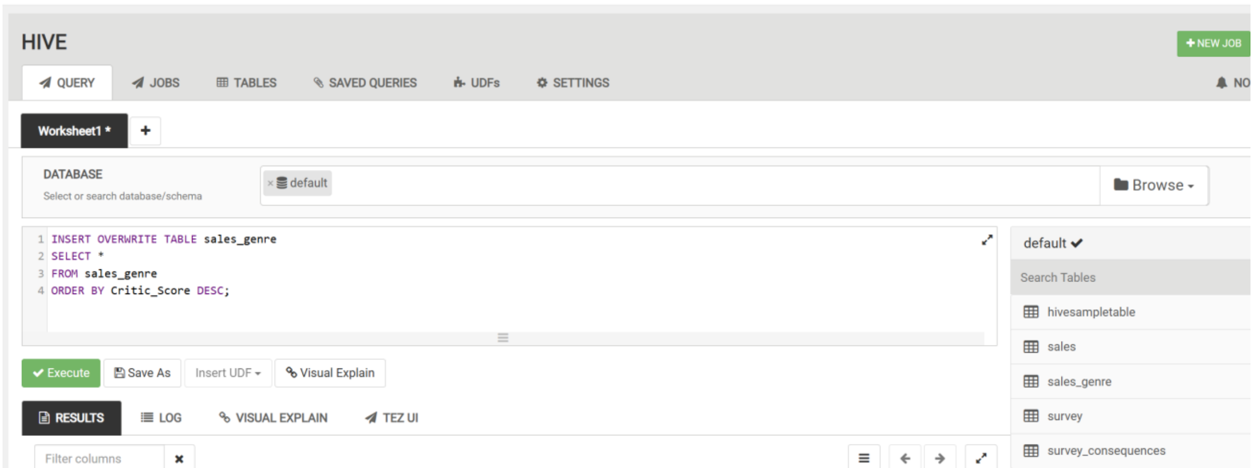
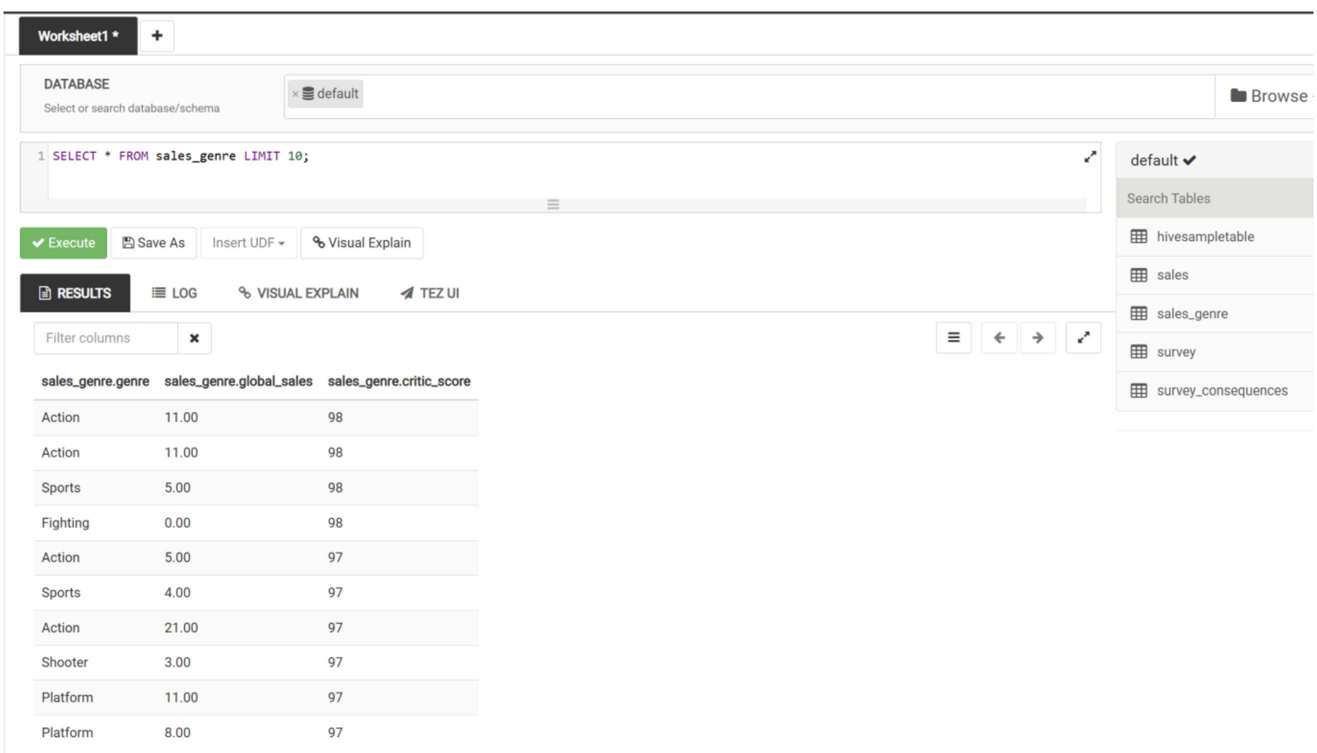
4)

INSERT OVERWRITE TABLE sales_genre
SELECT *
FROM sales_genre
ORDER BY Critic_Score DESC;



RESULTS FROM:
SELECT * FROM sales_genre LIMIT 10;



sales_genre.genre	sales_genre.global_sales	sales_genre.critic_score
Action	11.00	98
Action	11.00	98
Sports	5.00	98
Fighting	0.00	98
Action	5.00	97
Sports	4.00	97
Action	21.00	97
Shooter	3.00	97
Platform	11.00	97
Platform	8.00	97